Arsitektur Komunikasi SPI

Serial Peripheral Interface – Sintesis Dual-MCU untuk Sistem Akuisisi Data

SYSTEM ID CARD

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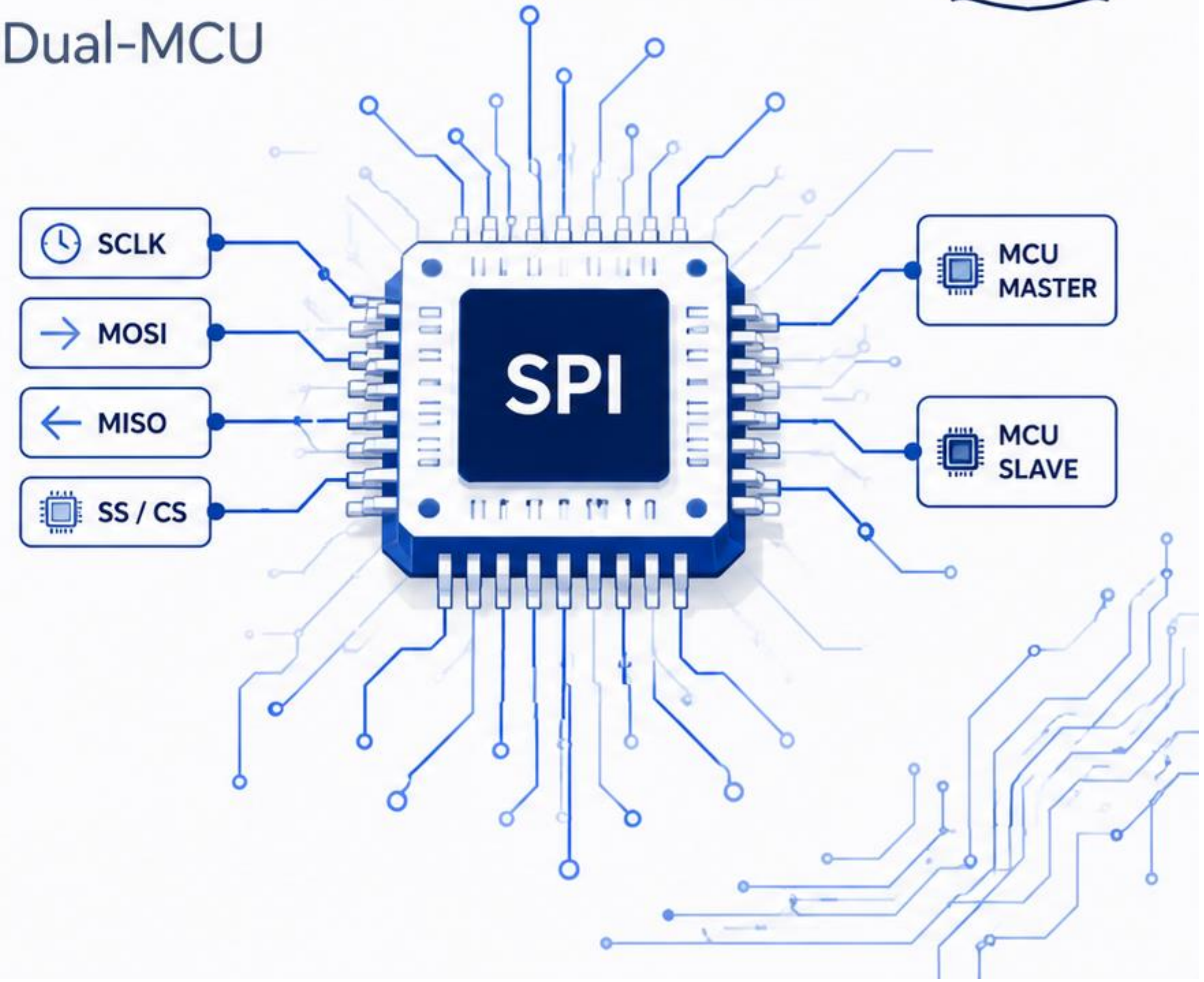


Tujuan Praktikum & Roadmap Sistem

TUJUAN UTAMA

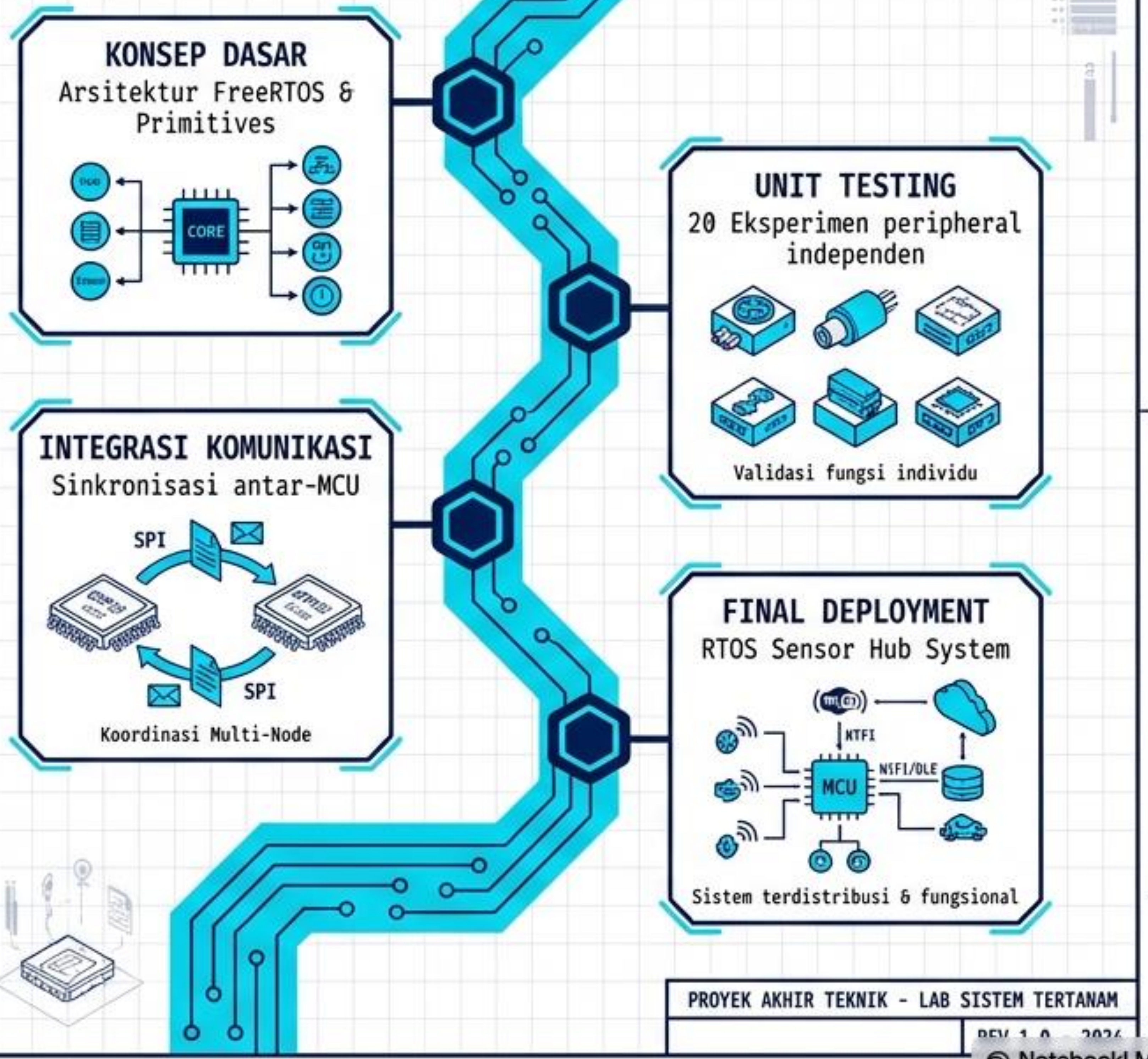
Penguasaan Task Scheduling & Context Switching.
Analisis siklus hidup tugas, manajemen prioritas, dan pemahaman mendalam tentang mekanisme context switching.

Implementasi komunikasi antar-task & ISR Handling.
Penerapan antrian, semaphore, mutex, dan penanganan Interrupt Service Routine (ISR) secara efisien.

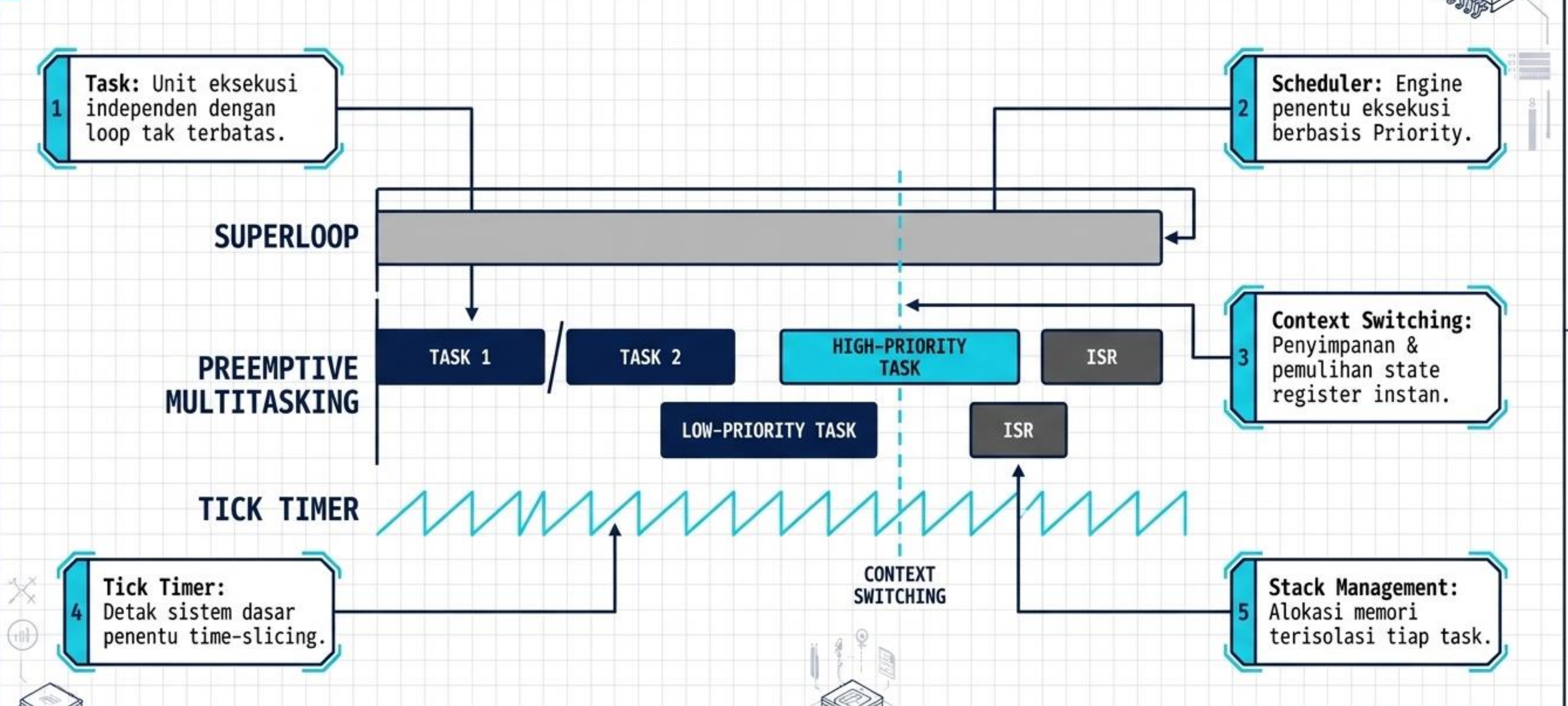
Integrasi Peripheral tingkat lanjut (ADC, DAC, I2C, SPI, Serial, Encoder).
Antarmuka ADC/DAC berkecepatan tinggi, komunikasi bus serial I2C/SPI, UART, dan pemrosesan encoder quadrature.

Sinkronisasi Multi-MCU (ESP32 & STM32) menggunakan RTOS.
Koordinasi tugas terdistribusi, protokol sinkronisasi, dan komunikasi data real-time antara ESP32 dan STM32.

ROADMAP SISTEM



KONSEP DASAR FREERTOS



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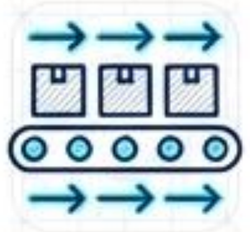
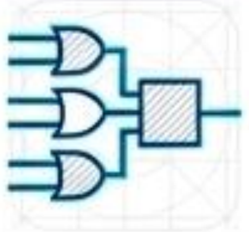
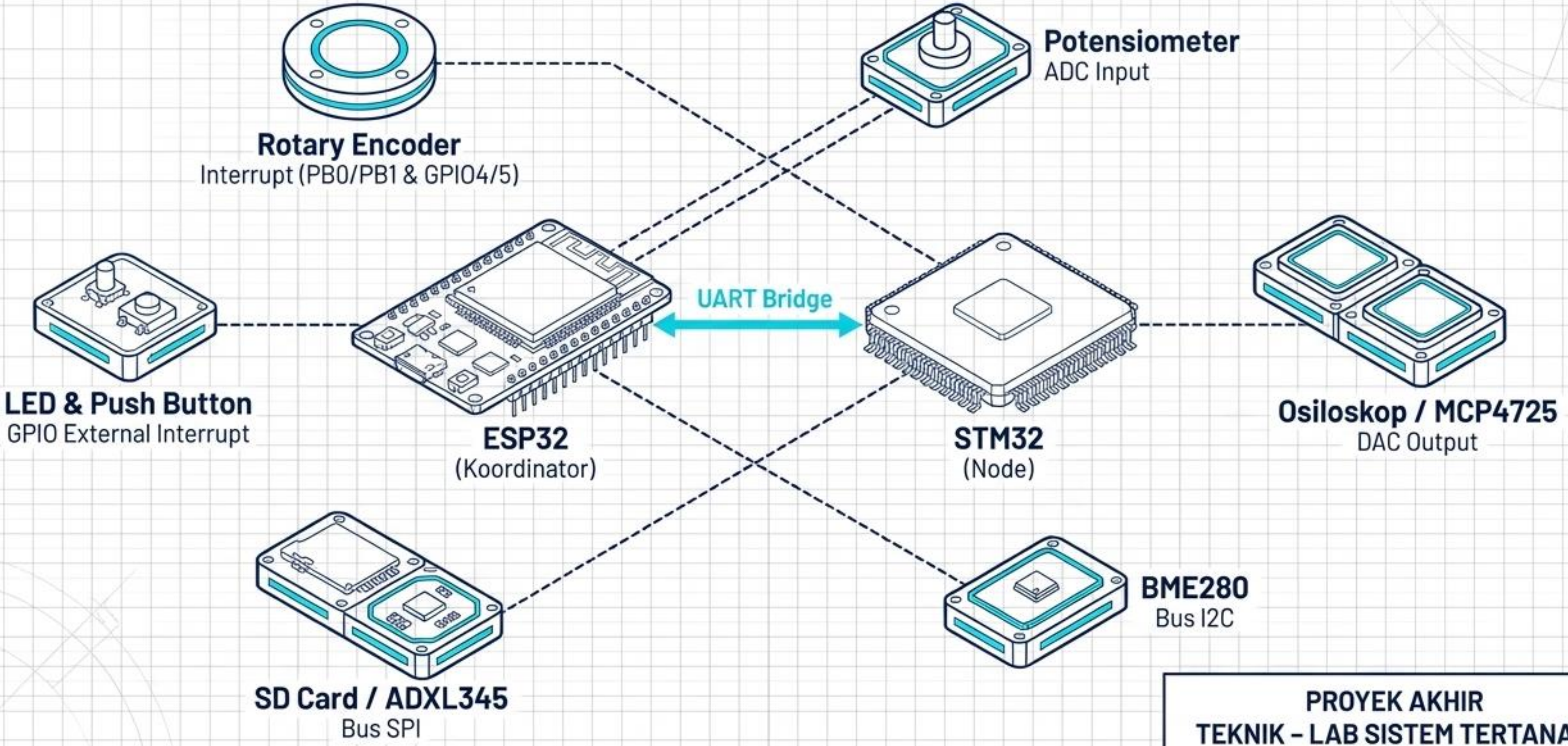
RTOS Primitives

Primitive	Ikon	Fungsi Utama	ISR-Safe API
Binary Semaphore		Signaling satu-arah (ISR ke Task)	`xSemaphoreGiveFromISR()`
Mutex		Proteksi Shared Resource (Priority Inheritance)	`Not for ISR`
Queue		Transfer data FIFO antar-task	`xQueueSendFromISR()`
Event Group		Sinkronisasi berbasis bitmask (Multiple Events)	`xEventGroupSetBitsFromISR()`

Aturan Emas: ISR harus sangat cepat. Jangan gunakan **vTaskDelay()** di dalam ISR.

HARDWARE & SISTEM PRAKTIKUM



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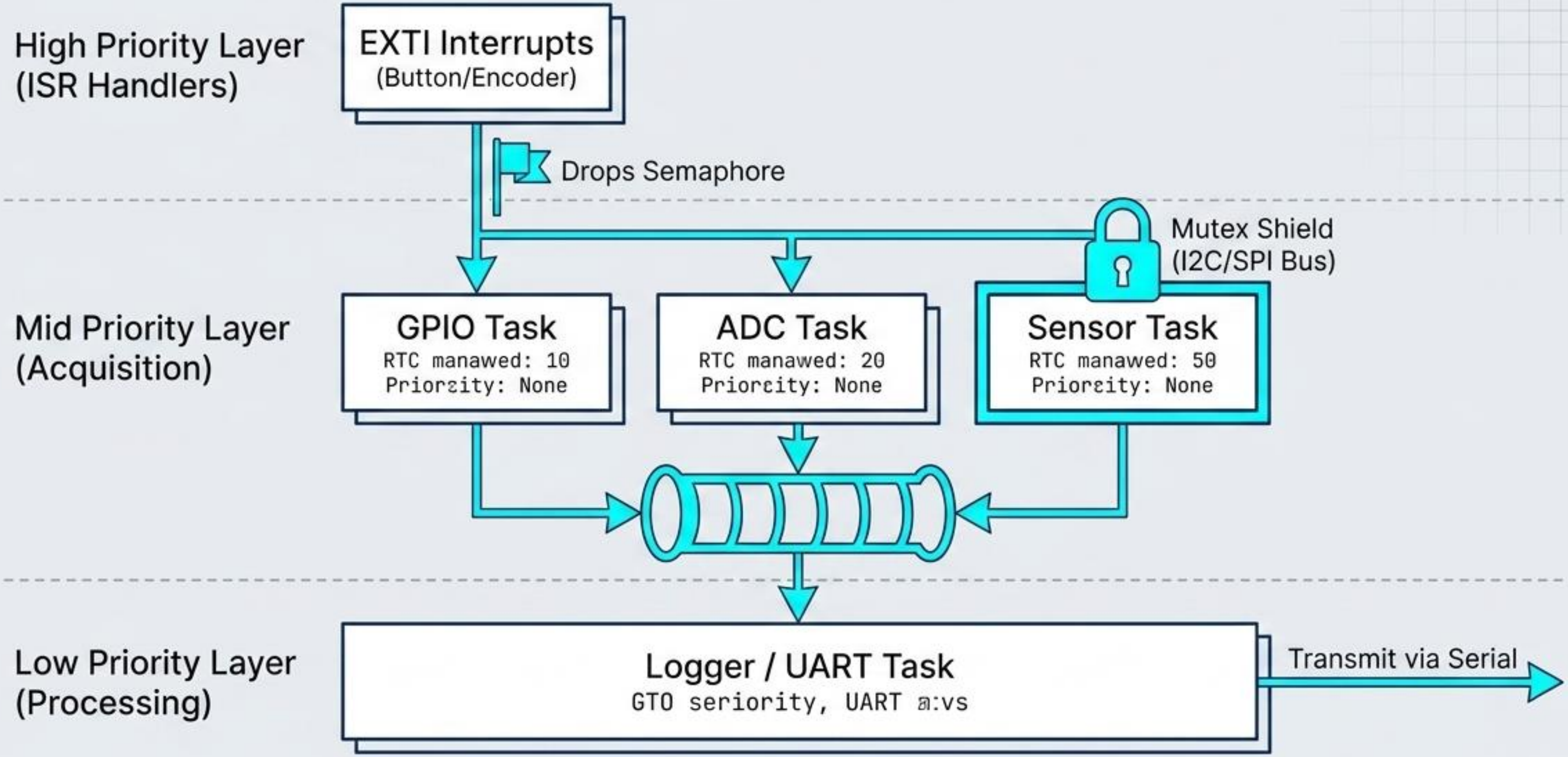
Arsitektur Multi-Tasking



STM32 Experiments (Unit Testing)

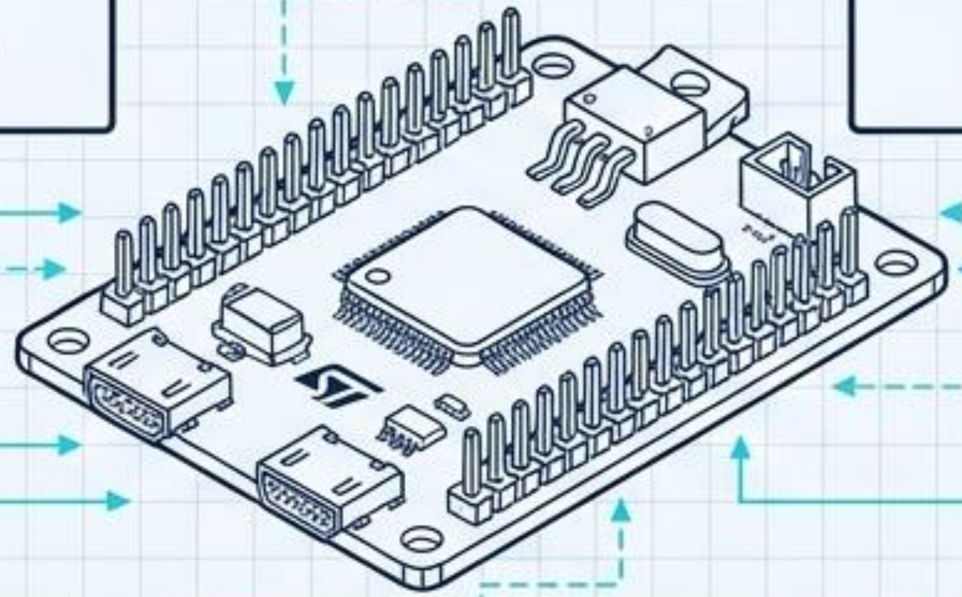
1. GPIO Blink RTOS
"Non-blocking osDelay"

2. Ext. Interrupt
"ISR ke Semaphore"

3. Encoder
"Queue event 2-Pin"

4. Serial UART
"RX Queue buffer"

5. DAC Task
"Waveform Lookup Table"



6. ADC Task
"100ms periodic read"

7. I2C RTOS
"Mutex bus protection"

8. SPI RTOS
"Mutex data transfer"

9. Multi-task
"ISR vs Polling sinkron"

10. System Sync
"Semaphore & Mutex stabil"

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ESP32 Experiments (Unit Testing)

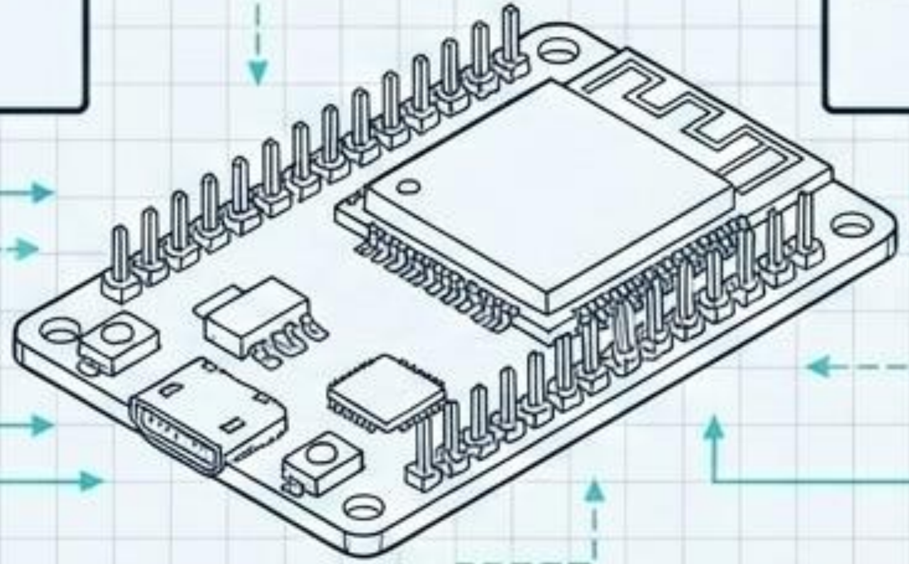
1. GPIO Blink
"Native vTaskDelay"

2. Ext. Interrupt
"GPIO falling edge ISR"

3. Encoder
Queue rotasi presisi

4. Serial UART
uart_driver_install event

5. DAC Task
"Kontrol frekuensi analog



6. ADC Task
Queue data tegangan

7. I2C RTOS
Thread-safe master config

8. SPI RTOS
Bus sharing thread-safe

9. Multi-task
Event Group flag state"

10. Queue Set
"Integrasi Event Group

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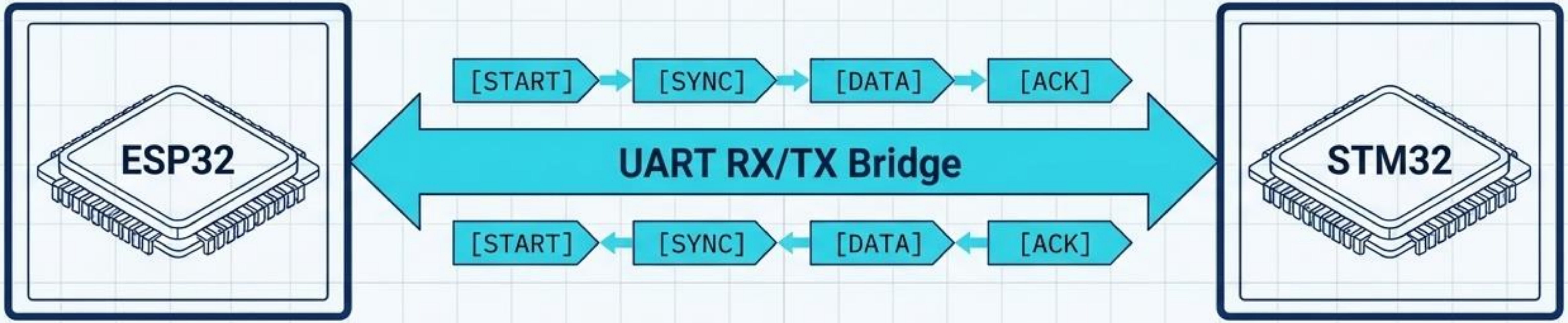
Multi MCU RTOS Communication



Task Synchronization

LED blink tersinkronisasi presisi.

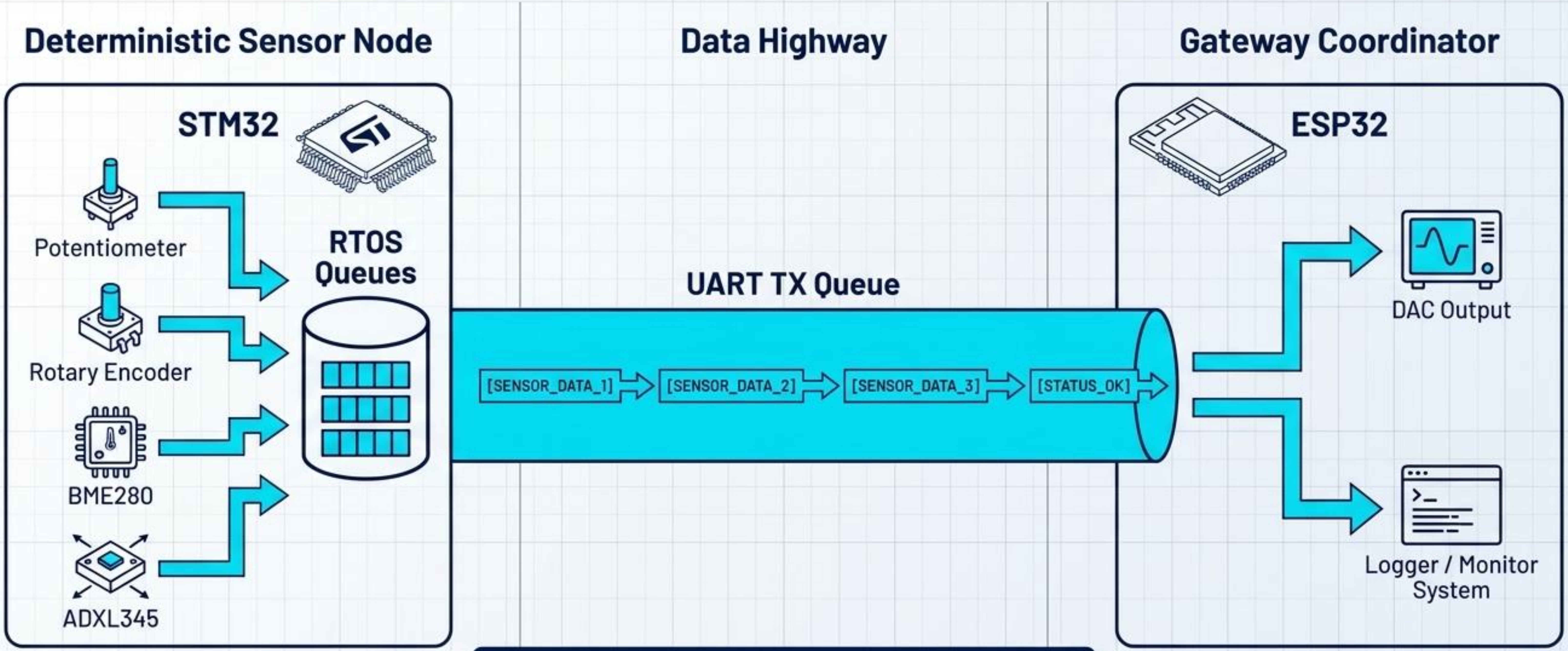
Data Exchange

ADC STM32 memicu output DAC ESP32 secara real-time.

Sensor Sharing

Akses pool sensor I2C/SPI bergantian tanpa bottleneck.

Project Final RTOS Sensor Hub



Stable Concurrent Execution

Penjelasan Program (Alur Eksekusi)

Initialization

Init Peripheral → Create Queue/Semaphore/Mutex → Start Scheduler.

↓

Acquisition

ISR tereksekusi → Give Semaphore → Task Take Semaphore → Ambil data.

↓

Protection

Akses I2C/SPI dikunci oleh Mutex → Data terbaca → Mutex dilepas.

↓

Transfer

Data dikemas → Dikirim via Queue → UART mentransmisikan data.

↓

Safety Net

Error handling & Retry mechanism aktif (Timeout tanpa memblokir OS).

Demonstrasi Praktikum (Hasil Validasi)

- ✓ LED blink RTOS berjalan stabil (1Hz).
- ✓ Interrupt Button merespons tanpa blocking di ISR.
- ✓ Counter Encoder presisi secara real-time (CW/CCW).
- ✓ Komunikasi UART Queue lancar (Echo/Command aktif).
- ✓ Output ADC dan DAC tervalidasi via osiloskop.
- ✓ I2C dan SPI beroperasi aman dengan proteksi Mutex.
- ✓ Multi-tasking terkelola tanpa Task Starvation.
- ✓ Sinkronisasi ESP32 dan STM32 berjalan sempurna.
- ✓ **RTOS Sensor Hub aktif secara penuh.**

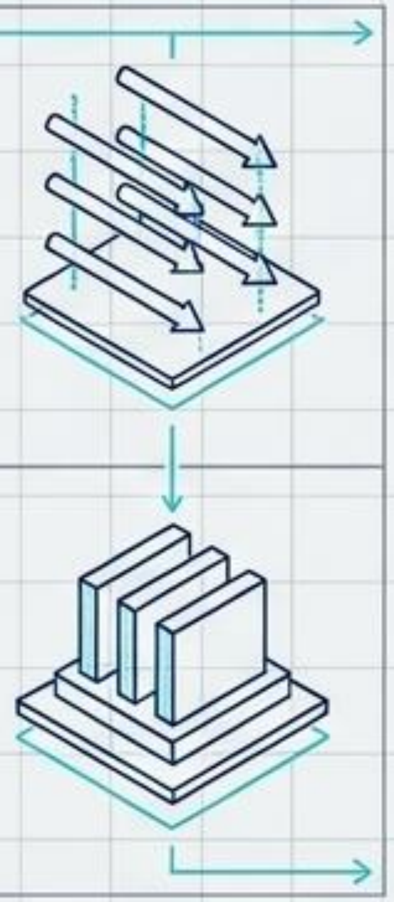
Kendala dan Troubleshooting

Diagnostics & Solutions	
Diagnosis Kendala	**Solusi Engineering**
Deadlock Mutex	*Enforce* pola pasangan take-release di task yang sama.
Stack Overflow	Gunakan **uxTaskGetStackHighWaterMark()** & kalibrasi config.
Race Condition	Terapkan **Mutex** pada Shared Resources (I2C/SPI/UART).
Interrupt Conflict	Hindari **vTaskDelay** di **ISR**; gunakan API khusus **FromISR**.
Queue Overflow	Tingkatkan buffer size atau prioritas task pembaca.

Kesimpulan

Stabilitas Konkuren

FreeRTOS sukses mengubah hambatan single-threaded (Superloop) menjadi arsitektur konkuren yang deterministik dan tangguh.

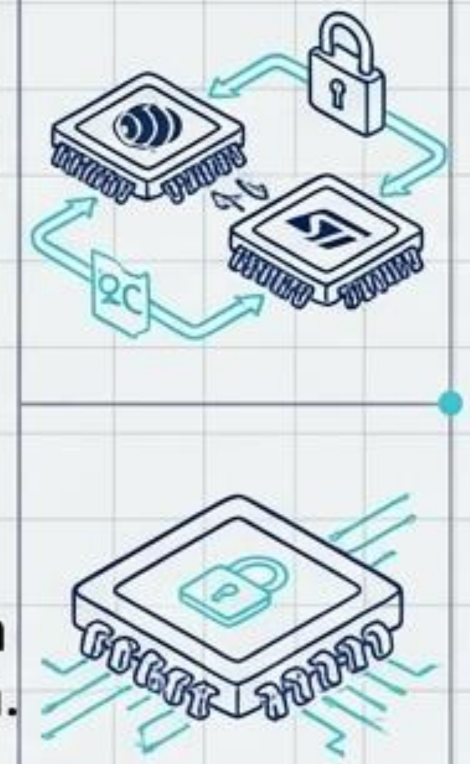


Sinkronisasi Presisi

Penggunaan Queue, Semaphore, dan Event Group memastikan data dari ISR dan task tersalurkan tanpa kehilangan informasi.

Integrasi Tanpa Konflik

Implementasi Mutex berhasil mengamankan resource I2C/SPI pada sistem ESP32 dan STM32 terpadu, menghilangkan Race Condition.



Terima Kasih.

FreeRTOS Middle | RTOS Sensor Hub System

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